

LETTERKENNY ARMY DEPOT
PURCHASE DESCRIPTION
COMPUTER NUMERICALLY CONTROLLED (CNC) 5 AXIS GANTRY MILL

1.0 General Information:

1.1 Scope: Request the purchase of one (1) CNC 5 Axis Gantry Mill. The gantry mill is capable of 5 axis simultaneous machining. This specification establishes the minimum requirements for the purchase and installation of contractor installed CNC 5 Axis Gantry Mill including utility, millwright, and foundation work at Letterkenny Army Depot (LEAD). The contractor shall provide all professional architectural, engineering and construction services for the preparation of all necessary drawings, specifications, calculations, cost estimates and related design, fabrication, construction, installation and testing services for the completion of this project. The gantry mill shall be installed turnkey in accordance with Original Equipment Manufacturer specification and standards. Specifications are based upon commercially available products and shall meet or exceed the minimum requirements specified in section 4.0 Specifications.

1.2 Place and Performance of Installation: All work shall be performed between the hours of 6:00 AM and 3:30 PM, Monday through Thursday and every other Friday 6:00 AM – 2:30 PM, exclusive of Federal Holidays and special depot closures (typically the Monday before or Thursday after Federal Holidays that fall on a Saturday, Sunday, Tuesday or Thursday). Other hours, weekend, and holiday work will be considered separately as requested by the contractor with adequate advanced notice and approved by the Contracting Officer.

1.3 Security Requirements: Contractor and all associated sub-contractor's employees shall comply with applicable installation, facility and area commander installation/facility access and local security policies and procedures (provided by Government representative). The contractor shall also provide all information required for background checks to meet installation access requirements to be accomplished by installation Provost Marshal Office, Director of Risk Management or Security Office. Contractor workforce shall comply with all personal identity verification requirements as directed by DOD, HQDA and/or local policy. Should the Force Protection Condition (FPCON) change, the Government may require changes in contractor security matters or processes. The contractor and all associated subcontractor employees shall also comply with adjudication standards and processes using the National Crime Information Center Interstate Identification Index (NCIC-III) and Terrorist Screening Database (TSDB) (Army Directive 2014-05/AR 190-13), applicable installation, facility and area commander installation/facility access and local security policies and procedures (provided by Government representative), or, at OCONUS locations, in accordance with status of forces agreements and other theater regulations.

1.3.1 Security, Safety and Fire Protection: All work shall comply with LEAD Regulation 385-1 and Pamphlet 385-4. The equipment and the installation shall also meet all the safety criteria set forth in OSHA 1926.501 and ANSI/ASSE Z359.

1.3.2 Media Devices: All pictures shall be taken by LEAD representative and approved before release.

1.3.3 AT LEVEL I Training: All contractor employees, including subcontractor employees, requiring access to Army installations, facilities, and controlled-access areas shall complete AT Level I awareness training within 30 calendar days after the contract start date or effective date of incorporation of this requirement into the contract, whichever is applicable. The contractor shall submit certificates of completion for each affected contractor employee and subcontractor employee, to the Technical Point of Contact (TPOC) or to the contracting officer, if a TPOC is not assigned, within 30 calendar days after completion of training by all employees and subcontractor personnel. AT Level I awareness training is available at the following website: <http://www.letterkenny.army.mil/ctrresources/>

1.3.4 Physical Security: The contractor shall safeguard all Government property provided for contractor use. At the close of each work period, Government facilities, equipment and materials shall be secured.

1.3.5 iWATCH Training: The contractor and all associated sub-contractors shall brief all employees on the local iWATCH program (training standards provided by the requiring activity ATO). This local developed training will be used to inform employees of the types of behavior to watch for and instruct employees to report suspicious activity to the Technical Point of Contact (TPOC). This training shall be completed within thirty (30) calendar days of contract award and within thirty (30) calendar days of new employees commencing performance. The results must be retained by the contractor and available for review by a Letterkenny POC if requested. iWATCH training is available at the following website: <http://www.letterkenny.army.mil/ctrresources/>

1.3.6 OPSEC Training: Per AR 530-1 Operations Security, the contractor employees must complete Level I OPSEC Awareness training. New employees must be trained within thirty (30) calendar days of their reporting for duty and annually thereafter. Level I OPSEC Awareness training is available at the following website: <http://www.letterkenny.army.mil/ctrresources/>

1.3.7 Threat Awareness and Reporting Program (TARP) Training: For all contractors with security clearances. Per AR 381-12 Threat Awareness and Reporting Program (TARP), contractor employees must receive annual TARP training by a CI agent or other trainer as specified in 2-4b.

2.0 Government Responsibilities:

2.1 Government Furnished Property and Materials: The Government will provide the facilities, equipment, materials and/or services listed below:

2.1.1 Facilities: The Government will provide clear access to all working areas necessary for the contractor to perform the installation outlined, during the specified hours listed in section 1.2.

2.1.2 Equipment: none

2.1.3 Materials: Materials necessary for post installation accuracy and quality tests.

2.1.4 Services: none

2.2 Site Preparation: Letterkenny Army Depot will provide a clear area for installation of the gantry mill.

2.3 Utilities: LEAD will provide adequate utilities for the installation and operation of the machines within 50' of the equipment disconnect. LEAD will also provide temporary utilities for the operation of equipment required for the installation of the industrial equipment.

2.4 Delivery and Storage: The Government will not be responsible for loading, unloading, or providing any extended storage, prior to installation, of materials or equipment as part of this contract.

3.0 Contractor Responsibilities: The contractor shall be responsible to inform all of its workers, direct or contracted, and ensure they follow the information within this specification and perform the required training prior to delivery of the equipment. The contractor shall also have a representative onsite to manage the project and be the point of contact (POC) to communicate and coordinate with Letterkenny's POC.

3.1 Post Processor: The contractor shall supply one post processor written for the machine in the latest release of MasterCam that LEAD uses, MasterCam 2026 or newer. The post shall be proven before delivery to LEAD. The contractor shall supply verification software for the machine, written in the latest release of MasterCam, MasterCam 2026 or newer, whichever LEAD uses. The contractor shall have the post processor proven out by the machine manufacture before the post is delivered to LEAD.

3.2 Project Schedule: A project schedule shall be submitted by the contractor, within 30 days after receipt of the purchase order, to include a timeline for material delivery, installation, testing and training.

3.3 Utility Requirements: The contractor shall provide certified utility requirements within 90 days after receipt of the purchase order.

3.4 Field Verification: It shall be the responsibility of the contractor to verify, in the field, all measurements, locations of structures and utilities, and any other physical conditions which would affect the progress, or quantity of work. Any errors by the contractor, due to omission of field verification, shall not be the basis for any increase in contract price. Where existing utilities or other obstructions interfere with the progress of the contract, the contractor shall provide all materials and labor to relocate all obstructions as required and approved by the Contracting Officer.

3.5 Foundation: The contractor shall be responsible for the designing and installation of a special foundation to meet the requirements of this specification. The contractor shall inspect the installation site and prepare certified drawings of the proposed foundation

including all calculations. The drawings shall be the responsibility of the contractor to develop and verify accuracy and design and shall be approved by LEAD prior to any work being performed on the depot. The contractor shall provide all materials, equipment, tools, and labor necessary for the removal and disposal of the existing slab, excavation, grading, and installation of the new foundation. All excavated material and other excess material shall be removed to a location approved by LEAD and disposed of by the Contractor in accordance with Federal, State and local regulations.

3.6 Utility Installation: Utilities are to be run from the disconnect to the machine via the method most appropriate to the design and function of the machine. The contractor shall provide all materials, equipment, tools, and labor necessary for the installation of all utilities.

3.7 Disposal of Materials And Equipment: All salvageable materials and equipment removed by the contractor shall be delivered to the Depot Property Disposal Yard, or other sites as directed by the Contracting Officer, or his representative and shall remain the property of the Government. Delivery shall be made during normal operating hours of the receiving activity. No Government materials or equipment shall be removed from the reservation by the contractor. All packing, packaging, and scrap material accumulated during the performance of the contract shall be disposed of properly by the contractor in a Government supplied container.

3.8 Dust Control: The contractor shall control the amount of dust generated during the installation of the system foundation and components and shall prevent the spread of dust to occupied portions of the building. Use of water will not be permitted when it will result in, or create hazardous or objectionable conditions such as ice, flooding or pollution.

3.9 Protection of Existing Structures And Equipment: The contractor shall take all necessary precautions to insure against damage to existing structures and equipment to remain in place.

3.10 Installation of New Equipment: The contractor shall be responsible for supplying all parts, material, equipment, tools, and labor required to install the equipment provided under this contract. This shall include off-loading equipment, moving it to the installation site, positioning, and installation. The contractor shall make all connections to existing utilities necessary for proper operation. The contractor shall remove and dispose all packaging and preservative material. Removed components shall not be taken from the Depot property. Removed components shall be placed in a location designated by the Technical Point of Contact (TPOC).

3.11 Wiring: All electrical work and materials shall be in compliance with the National Electrical Code and National Electrical Manufacturers Association. All electrical installations shall be inspected and accepted by a certified electrician. All work shall be subject to inspection by LEAD electricians.

3.12 Pipe: Schedule 40 (minimum) black steel pipe and all valves necessary shall be installed to provide the equipment with required adequate air and water supplies. Changes in compressed air circuit material can be altered with approval from LEAD.

3.13 Fluids: The contractor shall provide all necessary fluids and chemical additives required for the start-up and proper operation of the 5 axis gantry mill. The contractor shall supply LEAD with SDSs for all the necessary fluids within 30 days after award of contract.

3.14 Permits: The contractor shall be adequately licensed, insured, and bonded to perform and complete this installation of the CNC 5 axis gantry mill.

4.0 Requirements:

4.1 Machine Options AND Features: The CNC 5 Axis Gantry Mill shall have the following minimum options and minimum sizes:

80 Tool Magazine With Load/Unload Arm
Part Probe and Tool Setter
Chip-Conveyors with Coolant Management
At Least One Set of Electronic and One Set of Paper Manuals
Manual Control Hand Wheel

4.1.1 Documentation: One set (each) of paper and one (each) electronic of operator instruction manuals, maintenance service manuals, and parts listing manuals shall be provided for the furnished systems. These manuals shall be written in the English language.

4.1.2 Preservation, Packaging and Delivery: The contractor shall utilize standard commercial methods for preservation and packaging appropriate for each unit and acceptable to commercial carriers. As a minimum, all areas susceptible to damage from exposure to the elements shall be preserved and/or packed to prevent damage. The contractor shall be responsible for ensuring the equipment is delivered to LEAD in good condition.

4.1.3 Warranty: The contractor shall furnish with the equipment a standard commercial warranty or full one year warranty (parts, labor, shipping, and personnel time and travel at minimum), whichever is greater. The contractor shall provide commercial warranties for any subcontracted components that will be underwritten. The contractor shall supply all corresponding warranty contact names and phone numbers before the end of the project. The warranty shall be effective beginning after final acceptance of the equipment and not before that time.

4.1.4 Inspection and Acceptance: The equipment shall be examined for design, construction, materials, components, and workmanship to determine compliance with the contract and this specification. Any deviation from these requirements shall be cause for rejection.

4.2 Machine Control System:

4.2.1 Part Programming: The gantry mill shall have a Fuji Automatic Numerical Control (FANUC) type control. All machine functions shall be controlled and monitored through the operator's control panel/interface and shall be capable of being set through machine code input into the control. The controls shall also accept off-site programs delivered to the machine via network. Each control shall include, but not limited to, the following minimum features:

- Simultaneous 5 axis operations
- 19 inch, minimum, color flat screen LCD display
- Touch screen
- Local Area Network Connection (Ethernet)
- Emergency Shutdown Buttons
- ISO Standard G-Code compatible

4.2.2 Software Compatibility: The contractor shall ensure that their machine's control and control software are compatible with the latest version of MASTERCAM programming software that LEAD uses. The contractor must supply LEAD a post processor that has been debugged and proven. The contractor shall verify and prove out the post-processor to be fully functional (see paragraph 3.1).

4.2.3 Collision Avoidance: The 5 Axis Gantry Mill must be capable of collision avoidance with the adjacent overhead crane. The Mill shall use a non-contact sensor to determine the distance from an overhead crane (danger zone). The Mill will have the machine control be programmed that will stop the machine's X-axis travel until the crane is out of the danger zone of the machine. The machine will not emergency shutdown, also known emergency stop (e-stop).

4.3 Machine Features:

4.3.1 The 5 axis gantry mill shall possess the following features at a minimum:

- Accept Set Up Tools Up To 22 Inches Long
- Accept Fixtures and Risers up to 24 Inches
- Ballscrew and Rack and Pinion Drive Systems
- machining table with 60,000 lb minimum capacity
- Spindle Speed Of at least 0-12,500 Rpm
- Spindle Power Of at least 80hp
- Spindle Torque of at least 750 ft.-lbs.
- Linear encoders on X, Y, and Z axes
- Maximum tool diameter using all pockets, at least 4.9 inches
- Maximum tool weight of at least 44 lbs.

4.3.2 Automatic Lubrication: The system shall include a centralized lubrication system to automatically provide lubricant to critical components.

4.3.3 Power Requirements: All power requirements shall be 460V 3ph. If any requirement is not, the contractor shall supply all required transformers.

4.3.4 Machining Envelop: At Least 169 Inches Length X 123 Inches Width X 50 Inches High (Not Including Tooling And Fixturing)

4.3.5 Table Size: Must accommodate machining envelop and tooling and fixturing

4.3.6 Footprint: Maximum of 73 Feet Long X 27 Feet Wide

4.4 General Requirements:

4.4.1 Surface Coating: All surfaces to be coated shall be cleaned of all foreign matter. The equipment shall be painted not less than one coat of primer and one coat of finish color, or powder coated. All coatings processes and applications shall be IAW standard commercial practice and industry standards.

4.4.2 Lubrication: Means shall be provided to ensure adequate lubrication to all moving parts. All oil holes, grease fittings and filler caps shall be easily accessible.

4.4.3 Energy Efficiency: All equipment that directly consumes energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within industry.

4.5 Safety and Environmental Requirements:

4.5.1 Equipment Hazards: Hazards inside the 5-Axis Gantry Mill shall be guarded against unauthorized access. Clearly visible signs and symbols shall warn personnel in the area, and interlocks, safety mats and redundant deactivation systems shall guard against accidental access to the end effector and moving parts. Guards shall be easily removable to facilitate inspection, maintenance, or repair.

4.5.2 Compliance With Environmental Laws And Regulations: Contractor shall comply with all applicable federal, state, and local environmental laws, statutes, regulations, executive orders, permits, Army regulations (with supplements), as well as Major Subordinate Command (MSC) and installation regulations and policies. Contractor shall immediately report any conflicts between applicable federal, state, local environmental laws, statutes, executive orders, and provisions of Army Regulation 200-1, and any specifications within this contract to the Technical Point of Contact (TPOC), as well as to the Chief, Environmental Management Division (EMD).

4.5.3 Generation Of Hazardous Waste: Contractor shall abide by the guidelines stipulated in LEAD EMSP 21-17, Waste Management and shall assign all hazardous waste management responsibilities to the appointed ECD. Contractor shall contact the Chief of the EMD to obtain technical assistance from the installation's Hazardous Waste Program Manager who will assist the ECD with achieving and maintaining compliance with hazardous waste storage and disposal requirements. The EMD will provide hazardous waste drums for the

Contractor to containerize hazardous waste generated through the contracted work. Where liquid hazardous waste is to be stored, the Contractor shall provide appropriate secondary containment and ensure containers are properly bonded and grounded where the chemical waste requires it. In addition, the Contractor shall be responsible for maintaining an appropriate spill response and control kit at all times. All spills are reportable and shall be coordinated with the TPOC and the Chief of the EMD. Contractor shall accumulate hazardous waste prior to disposal shipment in a satellite accumulation point at or near the point of generation, or in a less-than-90-day site, in accordance with Federal, State, Army, and installation regulations or policies. The Government is responsible for removal of all hazardous waste. Contractor shall contact the Hazardous Waste Program Manager for assistance. The Contractor shall provide a 24-hour notification to the Hazardous Waste Program Manager or designee that hazardous waste containers are ready for pick up and turn in by the EMD. The EMD will remove and dispose of manifested hazardous waste generated by contract activities from the installation to an approved, off post, permitted hazardous waste disposal facility. The Contractor shall coordinate appropriately with the TPOC and the Hazardous Waste Program Manager the management of all hazardous waste to include profiling and final disposal. The Contractor shall perform a daily walk-through to ensure all hazardous material and all hazardous waste is properly secured and segregated by compatibility. Contractor shall reference section 5 of this checklist to ensure compliance with Federal, State and local laws and regulations.

- 4.5.4 Use Of Hazardous Materials: Contractor shall contact the Chief of the EMD to obtain technical assistance from the installation's Safety Division Chief who will assist the Contractor with achieving and maintaining compliance with hazardous materials management guidelines, to include requesting and obtaining an Hazardous Materials Approval Group (HMAG), labeling, storage, secondary containment, issue, use, and disposal requirements. Contractor shall submit to the TPOC or Chief of the EMD a hazardous material inventory for any materials to be used in the performance of the contracted work. The hazardous material inventory will be submitted 30 days prior to commencement of work for contracts that exceed 180 consecutive days and 15 days prior to commencement of work for contracts requiring under 180 days of consecutive work. The inventory list shall contain the hazardous material type and maximum quantities of materials anticipated to be used and stored on-site. The hazardous material shall be properly identified and labeled to include any applicable identification number, such as National Stock Number or Special Item Number and include any applicable hazardous materials management guidelines found on EEJ-010, Hazardous Materials Management; LEADR 385-1, Chapter 10 and all applicable hazardous communication (hazcom) requirements or as required. Contractor shall ensure all hazardous materials are properly secured and segregated by compatibility and kept under his or her control at all times. No expired hazardous materials shall be used or stored at the installation. Where applicable, and prior to bringing or removing hazardous materials on, at, or from the facilities where the contracted work is to be performed, the Contractor shall submit copies, preferably in electronic form, of Global Harmonization System (GHS) compliant

Safety Data Sheets (SDSs) for all hazardous materials used and stored on-site during performance of the contract. Contractor shall not supply or deliver any hazardous materials or chemicals to an AMC installation that are listed on EPA toxic chemical list without prior written approval from Chief of the EMD. During the performance of the contracted work, the Contractor shall ensure that hazardous material containers remain under the control of the user at all times and that containers are properly closed when the material is not in use. The Contractor shall not bring into a Government installation any non-approved hazardous materials for use in the performance of their work. Where liquid materials are to be stored, the Contractor shall provide appropriate secondary containment. In addition, the Contractor shall be responsible for maintaining an appropriate spill control kit at all times. All spills are reportable and shall be coordinated with the TPOC and the Chief of the EMD. The Contractor shall, upon completion of each day's work, clean, remove, and properly containerize or dispose of any resultant hazardous waste generated by the work performed under this contract, where required by the installation. Upon completion of the contracted work, the Contractor shall conduct final clean-up of the work site, containerize any hazardous material/waste debris, and coordinate with the EMD for pick up and turn in for disposal. Any unused hazardous material and any non-ozone depleting substances or refrigerants brought in by the Contractor shall be recovered and removed from the installation by the Contractor prior to vacating the work site.

- 4.5.5 Prevention Of Spills: The Contractor shall follow LEAD's Integrated Contingency Plan (ICP) which includes the Spill Prevention, Control, and Counter Measure Plan (SPCCP) Plan and reference the LEAD SPCCP and ICP if transporting, processing, storing, or in any way managing hazardous waste, hazardous material, petroleum-oils-lubricants, or other restricted items. The Contractor shall ensure secondary containment is in place any time liquid materials are stored on-site and shall provide his or her own spill response supplies and have them readily available at each work site. In case of a spill, the person in control of the spill site, or their designated representative, shall take appropriate action to protect workers and bystanders; contain the spill (if it can be done safely); secure the spill site; restrict ignition sources; and immediately contact the installation Fire and Emergency Services (Fire Department) by dialing 911, the TPOC and the Chief of the EMD. Contractors storing 1,320 gallons or more of any oil-based product in an aboveground storage tank at a construction site shall follow LEAD's ICP. The Contractor shall immediately respond to actual emergencies and accidents, prevent or mitigate associated adverse environmental impacts, and contact the installation's Fire and Emergencies Services.
- 4.5.6 Corrective Action For Noncompliance: Contractor shall take immediate corrective action when given a verbal or written notice of environmental noncompliance or nonconformance by the TPOC or the Chief of the EMD. Failure or refusal by the Contractor to comply promptly may be grounds for the Contracting Officer to invoke the appropriate contractual remedies. This may cause all or part of the work to be stopped immediately until satisfactory corrective action has been taken by the Contractor.

- 4.5.7 Inspections Of Work Sites: Contractor shall submit to potential Federal, State, Army and installation work site environmental regulatory inspections and/or investigations into noncompliance and fully cooperate with such inspections/investigations by providing the appropriate records and documentation. Environmental regulatory agencies are authorized by law to inspect any work site for environmental compliance with regulatory requirements. If an inspection is conducted, it will not stop or disrupt ongoing contract activities. The inspection will only require the work site environmental officer, or supervisor/manager to answer questions and/or escort the inspector to specific work site areas with the potential to affect environmental quality. Typical environmental work site inspections are conducted in less than 30 minutes with an approximate frequency of one inspection every two months.
- 4.5.8 The contractor is reminded that LEAD maintains a Safety Management System (SMS) and is ISO 45001 certified, AS9100 and AS9110 certified. LEAD's safety policy and safety program (LEAD-R 385-1 and Safety Bulletins) can be found at <https://letterkenny.army.mil/visitors-Information>. It is the contractor's responsibility to be familiar with these documents and to comply with their contents.
- a. Contractor/Subcontractor will conduct work on Letterkenny Army Depot in compliance with the most current version of 29 CFR 1910 and 29 CFR 1926 (where applicable). Each contractor/subcontractor will be familiar with the types of work they are expected to perform and each particular phase of the operation and associated hazards. All worksites will be free of recognized hazards (per LEAD-R 385-1 27-2).
 - i. Each worksite will be assessed for hazards prior to the start of work. All training certifications and an Activity Hazard Analysis (AHA) will be on hand at the job site.
 - ii. Any and all on site work at LEAD shall be completed in accordance with applicable OSHA, DoD, Army, and LEAD Safety requirements. As a federal installation, additional requirements other than OSHA apply to contractors performing work at LEAD. Examples include: full conformance to all applicable NFPA standards, ANSI standards, and other consensus standards for the industry covered by the subject contract. It is the responsibility of the contractor to be familiar with all applicable standards prior to starting work. Additionally, the contractors shall make themselves familiar with and comply with LEAD's local safety regulation LEAD-R 385-1, AR 385-10, and LEAD-R 420-13 prior to the start of any work at LEAD.
 - iii. As part of the applicability of engineering controls, all equipment with rotating and moving parts at point of operation shall comply with LEAD 385-1 Chapter 17-3 machine guarding requirements.
 - iv. In case of a conflict among these laws and regulations, the most stringent law or regulation shall apply. It is the responsibility of the contractor to be familiar with all applicable standards prior to starting work.

- v. A list of all hazardous materials will be maintained at all times. All Safety Data Sheets (SDS) will be present and maintained at all worksites. A copy of all SDS's will be provided to the Safety Office at the beginning of all projects.
- b. The Contractor shall ensure all employees and all associated sub-contractor's employees possess all required licenses for operating any equipment used in the performance of this contract. The Contractor must ensure that any personnel performing under a contract are fully trained, licensed, certified, and qualified for the position in which they shall be serving.
- c. All contractor personnel shall wear the appropriate PPE when in the following work areas: industrial buildings, at construction sites, test sites and test facilities. Failure to wear/use PPE may cause removal from the job site and/or seizure of non-compliant equipment/items.
- d. If contractor personnel determine a procedure is unsafe, they shall discontinue work immediately and contact their Program Manager who shall notify safety personnel and the TPOC. As soon as LEAD Safety Office personnel determine that the unsafe conditions have been corrected, the TPOC will notify the contractor Program Manager that personnel may resume work.
- e. The contractor shall comply with reporting requirements of 29 CFR Part 1904 and AR 385-40 and APGR and shall report accidents resulting in injury or death and accidents resulting in damage to Government property, supplies, and equipment to the TPOC promptly of the occurrence. In case of injury/accident, the contractor shall complete an injury/accident investigation report and submit it to the TPOC with a copy to the LEAD Safety Office within 24 hours of the event, regardless of if the event is reported to another Safety Management System (SMS).
- f. All safety messages or recalls related to the safe use of products provided by the contractor will be provided to the LEAD POC for the Contract within 5 days of contractor's awareness of the message or notice.

4.6 Installation Requirements:

4.6.1 Design and Installation: The contractor shall provide all parts, labor, and services for the design, fabrication and installation of the gantry mill and associated equipment. The contractor shall provide competent English speaking personnel to place the system in operation and to conduct the acceptance tests and training of depot personnel. The contractor, through his service personnel, shall be fully responsible for the offloading, installation, utility connection, activation, and testing of the system and the service personnel shall remain until system activation and testing has successfully been completed. Under no circumstances should the contractor plan on gratuitous Depot assistance during installation, activation, and testing such as diagnosis, troubleshooting, or repair of the system.

4.6.2 Badges: Identification badges will be issued for admittance of personnel before performing work or service on this contract. The identification badge is the property of the US Government and must be returned upon termination or demand. The contractor/vendor agrees that he will ensure all badges issued to employees are

returned to the Badge and Identification Section, Security Division, Letterkenny Army Depot promptly on termination of need or on expiration, which occurs first. The badge room is located in Bldg 2, phone # 717-267-5301. Hours of operation are 7:30 AM to 3:30 PM, Monday thru Friday, exclusive of Federal Holidays.

4.6.3 Fire Extinguishers: The LEAD R 385-1 designates the type and number of fire extinguishers required to perform the contracted work. The LEAD Fire Department will not provide any extinguisher support. The contractor can gain the above information through a pre-construction conference, visiting the fire department, or phone 717-267-8231.

4.6.4 Use of Portable Gas or Arc Equipment: The contractor shall obtain a permit, from the LEAD Fire Department, to use portable gas or arc equipment for cutting, welding or open flame operations. The Fire Department will require one day advance notice to permit necessary inspection before actually starting work, and notification when work is finished to allow for re-inspection. The Fire Department can be notified by calling ext. 8231. The contractor shall take necessary precautions to prevent fires IAW NFPA Standard No. 51B and LEAD R 420-13.

4.6.5 Spills: Any spills or releases of materials or wastes must be reported immediately to the site supervisor, the LEAD Fire Department, and the LEAD Environmental Office per the LEAD Installation spill Contingency Plan (ISCP). Any such spills or releases generated by the contractor shall be contained, cleaned and disposed of by the contractor per the direction of the LEAD on-scene coordinator.

5.0 Applicable Documents:

OSHA Safety and Health Standards
29 CFR 1910 Occupational Safety and Health Standards
(copies may be obtained at <http://www.osha.gov>)

Letterkenny Army Depot (LEAD) Publications:
LEAD Regulation 385-1: LEAD Safety and Occupational Health Program
LEAD Pamphlet 385-4 – Safety Requirements for Security, Safety, and Fire Prevention for Contractors Performing Work on LEAD

National Fire Protection Association
NFPA Standard 70 – National Electric Code
NFPA Standard 79 – Electrical Standard for Industrial Machinery
(copies may be obtained from <http://www.nfpa.org/>)

6.0 Quality Assurance:

6.1 Quality Conformance Inspection: The 5 axis CNC gantry mill components shall be subject to a quality conformance inspection, performed by a designated Depot representative, to determine compliance with this specification.

6.2 Tests: The contractor shall be responsible for developing test procedures to demonstrate compliance with the requirements of this specification. All test procedures

shall be submitted to and approved by Depot Technical Point of Contact (TPOC) prior to the start of testing. To support the testing procedures, LEAD will provide a supply of material for use during the performance tests and training. The material may include mild steel, aluminum, stainless steel and non-ferrous materials of varying thicknesses, depending upon availability. The contractor shall run, successfully, a proven part program through their post-processor and download to the machine. The part shall be run from their download and checked for accuracy of the part and program. The contractor shall prove out all problems with the post.

6.3 Training: The contractor shall be responsible for providing training of four (4) Depot personnel in the operation of the 5 Axis MCs and software and four (4) Depot personnel in the maintenance of the equipment supplied under this contract. All instructions, oral and written, shall be in the English language. The project LEAD TPOC will coordinate with individuals to be trained upon successful completion of all installation and test specifications.

7.0 Information Technology (IT) Industrial System Requirements:

7.1 Operating System (OS) Specifications:

7.1.1 If applicable, the system's OS must be Microsoft Windows 10, Windows Server 2012R2, or Red Hat Linux v6.X. Any OS supplied with the system must contain all service packs, updated patches, and hot fixes. All patching and fixes will be current as of a predefined date agreed upon by the vendor and the Government.

7.1.2 Defense Information System Agency (DISA) Security Technical Implementation Guides (STIGS) are configuration standards for the Department of Defense and contain technical guidance to lock down information systems and software applications that might otherwise be vulnerable to malicious attacks. The contractor shall be responsible for ensuring all appropriate STIGs are implemented and their software runs acceptably while secure.

7.1.3 The Army Golden Master (AGM) is a collection of security settings for Microsoft Operating systems which must be applied to all Army computer systems. LEAD will supply the AGM to the contractor who shall ensure their software functions as intended, when installed on top of the AGM.

7.2 **Programmable Logic Controllers (PLC):** If a PLC is to be installed within the system, it shall be manufactured by a Government approved source and shall provide an Ethernet adapter to allow for connection to the Industrial Network if required. If an Ethernet connection is used to interconnect the PLC to peripheral equipment, an additional Ethernet port shall be provided.

7.3 System Software Specifications:

7.3.1 **Software Installations:** If installing application code at LEAD, the contractor shall install the code within a standard container (E.g., Program Files) and located in a volume separate from the standard System Files. Interactive applications are

required to run as services and daemons and any time WEB services are used, file types (E.g., XML and HTML) must be separated.

7.3.2 Software Licensing: When installed, the software licensing for computers, control units and other IT systems will be surrendered to the Information Technology Configuration Manager and will become the property of the Government upon acceptance of the system.

7.3.3 If software has been customized for LEAD's use or is any software other than Commercial Off The Shelf (COTS); the contractor shall provide copies of all source code, firmware and software used in the system, as well as one set of backup and configuration software for PLC's, control and communication computers, and data acquisition. All programs shall be tested and fully operational before acceptance. All acceptance testing will adhere to common User Acceptance Testing (UAT) standards as defined by the Government.

7.3.4 If any special programming software or programming software interface devices such as hardware "keys", dongles or "PIC" modules, are required to access software, these programs and devices shall be provided to LEAD along with the necessary registration information. If custom application software was written and provided, a copy of the required compiler and compiler license shall also be provided. At acceptance, all items shall become property of the Government. Software and hardware purchased for this project shall be registered to Commander, Letterkenny Army Depot, Chambersburg, PA 17201.

7.4 System Environment Specifications:

7.4.1 All supplied software must be capable of operating in a multi-user environment and data which contains permissions information shall be hidden from view and inaccessible to those with less than administrator level access. Shared accounts shall not be used, and users shall only be authorized the minimum security rights to operate the application software. Users will not have access to system or software configuration items (e.g., Control Panel, software and hardware installation). Under no circumstances will the intended end user log in using the Administrator or Root account for routine operation of the equipment. The use of these accounts will be restricted to only administrative activities which will be carried out by LEAD Information Technology personnel. All software products shall employ a minimum of three permission based levels of access.

7.4.1.1 Administrator - Full control at root level, restricted to LEAD IT personnel and when required, the Contractor.

7.4.1.2 Maintenance - Mid-level control required to perform maintenance and alter control parameters as needed.

7.4.1.3 Operator - Minimum control necessary to perform the routine operation of software. The Operator shall not be able to alter the programming of software, make changes to executable files, or modify control parameters.

- 7.4.2 Application Interoperability: If an application will interfere with the operation of the industrial system, E.g., (McAfee Anti-Virus), the Contractor shall provide a written statement to the LEAD IT Directorate explaining the technical restrictions placed upon them.
- 7.4.3 Auditing: If applicable, the contractor shall enable audit files and data logs for access by LEAD Cybersecurity personnel to audit access and actions on the equipment.
- 7.4.4 User Acceptance Testing (UAT): LEAD Cyber Security Personnel will have the ability to conduct validation testing with approved security tools prior to equipment being introduced to the shop floor. This testing will also satisfy any UAT requirements. A UAT Plan will be furnished prior to testing and all equipment documentation will be made available for review during the test. If discrepancies are discovered the contractor shall remediate identified issues prior to Government acceptance.
- 7.5 Documentation Requirements:
- 7.5.1 A complete set of instruction manuals containing all of the information necessary to operate, maintain, and reinstall all software and hardware contained within the system shall be provided by the Contractor. These manuals shall include step-by-step instructions which facilitate the rebuilding of the system in the event of a fatal system crash.
- 7.5.2 If networked, the Contractor shall provide the system's network design drawings to the Information Technology Directorate. These drawings should include complete directional data flow diagrams that include all ports and protocols in use.
- 7.5.3 The Contractor shall provide to Industrial Automation at least two electronic copies of all control unit programs. These copies should include description files of the ladder logic and any cabling required to connect to the control systems. No control units will be password protected and no other form of lockout measure shall be implemented by anyone other than LEAD IT employees. The Contractor shall also provide electrical wiring diagrams for all installed control systems to this LEAD Industrial Automation group.
- 7.5.4 Certificate of Networkiness: Certificate of Networkiness (CoN) certification ensures Automated Information Systems (AIS) utilizing the Army Network are developed in compliance with the Clinger Cohen Act and are secure, supportable, sustainable, and compatible with the Army Enterprise Infrastructure (AEI) (as defined in AR 25-1). All new AIS capabilities and all capability modifications and upgrades must be assessed to validate their Networkiness (network security, network impact, compatibility with the infrastructure, infrastructure requirements, spectrum support, security policy compliance, Joint Technical Architecture –Army (JTA-A) standards compliance, communications and information manpower, training, logistics support, schedule, and funding). If software is installed within the

system, the Contractor shall work with LEAD Directorate of Information Management (DOIM) personnel to check for CoN availability and when not available, ensure a CoN is requested and all documentation is submitted.

7.6 Proposal Deviations:

7.6.1 Proposals or specifications that deviate from those described herein shall be coordinated for specific approval or rejection by the LEAD Information Technology representatives prior to shipment, installation, or acceptance of the equipment.

8.0 Final Acceptance: Once all conditions of this contract are met, final acceptance will be after 100 hours of continuous operation without a breakdown or failure, or 14 working days, whichever comes first. If the machine has a failure or breakdown and it is determined that it is the fault of the machine hardware or software, the operating time and 14 days timeframe start over each time this happens. If the failure or breakdown is determined to not be the fault of the machine hardware or software, the 100 hour/14 day time continues. After the 100 hour operating time or 14 days have been met, final payment of this contract will be made.